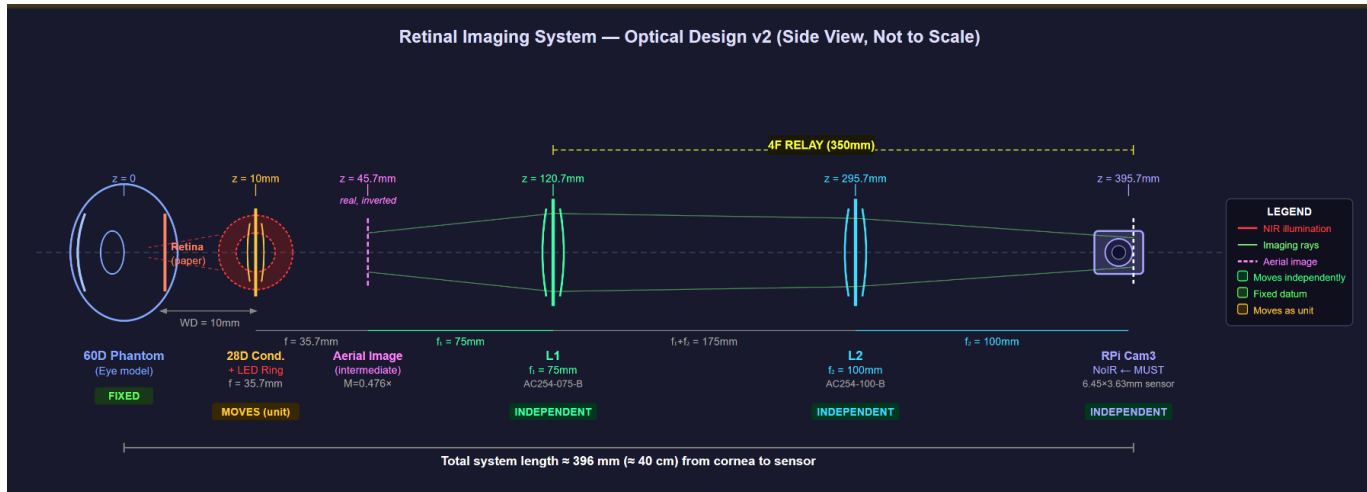


Retinal Imaging System

Optical Design v2 — Technical Specification

Capstone Project — March 2026



1. System Overview

This document presents the full optical design for a non-mydiatic retinal imaging system using the eye as its primary objective lens (indirect ophthalmoscopy principle), a condensing lens to form an intermediate aerial image, and a 4F relay system to project that image onto a Raspberry Pi Camera Module 3 NoIR sensor.

The design incorporates annular NIR LED illumination to avoid corneal specular reflection, and is compatible with an eye phantom for bench testing. All calculations below use the Gullstrand simplified model eye as the basis for the eye's optical parameters.

1.1 Optical Train at a Glance

Element	Position (from cornea)	Key Parameter	Notes
Cornea / Eye	$z = 0$ mm	$f = 17$ mm (air)	Gullstrand model; axial length 24 mm
Condensing Lens + LED Ring	$z = 10$ mm	$f = 35.7$ mm (+28 D)	Working distance 10 mm from cornea; LED ring surrounds aperture
Aerial Image Plane	$z = 45.7$ mm	— (intermediate)	Inverted real image of retina; $M = 0.476\times$ relative to retina
4F Lens L1	$z = 120.7$ mm	$f_1 = 75$ mm	Thorlabs AC254-075-B achromat (NIR B-coat)
4F Lens L2	$z = 295.7$ mm	$f_2 = 100$ mm	Thorlabs AC254-100-B achromat (NIR B-coat)

RPi Cam3 NoIR Sensor	$z = 395.7 \text{ mm}$	$4608 \times 2592 \text{ px}, 1.4 \mu\text{m}$	Total system length $\approx 40 \text{ cm}$ from cornea
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2. Sensor Specification (RPi Camera Module 3 NoIR)

The Raspberry Pi Camera Module 3 NoIR variant is required. The standard variant includes an integrated IR cut filter that blocks wavelengths above $\sim 700 \text{ nm}$, which would eliminate the NIR illumination entirely. The NoIR variant removes this filter, giving full sensitivity across the NIR range used for retinal imaging.

Parameter	Value
Sensor	Sony IMX708 (back-illuminated stacked CMOS)
Resolution	4608×2592 pixels (11.9 MP)
Pixel size	$1.4 \mu\text{m} \times 1.4 \mu\text{m}$
Sensor dimensions	$6.45 \text{ mm} \times 3.63 \text{ mm}$ (7.40 mm diagonal — verified against datasheet)
IR cut filter	ABSENT (NoIR variant) — required for 850 nm illumination
Autofocus	Phase Detection AF (PDAF) — set to manual infinity via libcamera
Interface	CSI-2, $15 \times 1 \text{ mm}$ FPC connector, 200 mm ribbon

3. The Eye as the Primary Objective Lens

In this indirect ophthalmoscopy configuration, the eye itself acts as the objective lens. Parallel light from the illumination ring enters the eye, scatters off the retina, and exits the eye as a diverging wavefront. A condensing lens placed at the working distance from the cornea collects this light and forms a real, inverted aerial image of the retinal plane.

Eye Parameter	Value / Justification
Eye model	Gullstrand simplified schematic eye
Posterior focal length (in vitreous)	17.0 mm ($n_{\text{vitreous}} = 1.336$)
Anterior focal length (in air)	$\sim 17.0 \text{ mm}$ (used for magnification formula)
Axial length	24 mm (cornea to retina)
Total refractive power	$\sim 78.6 \text{ D}$
Pupil diameter (dilated)	$6\text{--}8 \text{ mm}$ (phantom: fixed at $\sim 6 \text{ mm}$)
Note on myopia/hyperopia	Myopia/hyperopia shift the axial length. Adjust condensing lens WD accordingly, or use liquid lens to compensate (see Section 8)

Key insight from the whiteboard: the standard thin-lens equation does not apply directly because the image forms on the retina inside the eye (a medium of $n=1.336$). When computing the aerial image, we

use the exit-pupil formulation — for an emmetropic eye, light exits parallel (retina at focal plane), and the condensing lens then forms the aerial image at exactly its back focal length.

4. Condensing Lens & Aerial Image

4.1 Lens Choice: +28 D

A +28 D condensing lens (equivalent to those used in clinical indirect ophthalmoscopy) provides a good balance between working distance and field of view. Compared to +20 D, it gives a shorter working distance but wider retinal field coverage.

Parameter	Value
Power	+28 D
Focal length	$f = 1000/28 = 35.71 \text{ mm}$
Working distance from cornea	10 mm (standard clinical WD for +28D)
Aerial image location	35.71 mm behind condensing lens = 45.71 mm from cornea
Magnification (eye + condensing)	$M = f_{\text{eye}} / f_{\text{cond}} = 17.0 / 35.71 = 0.476\times$ (inverted)
Aerial image of macula (5.5 mm retinal dia)	$5.5 \times 0.476 = 2.62 \text{ mm diameter}$
Aerial image of full retina (~32 mm)	$32 \times 0.476 = 15.2 \text{ mm diameter}$
Suggested component	Volk 28D aspheric condensing lens (clinical grade)

5. 4F Relay System Design

The 4F relay consists of two lenses (L1 and L2) separated by the sum of their focal lengths ($f_1 + f_2$). The aerial image is placed at the front focal plane of L1, and the sensor is placed at the back focal plane of L2. This arrangement is telecentric in image space and provides well-defined, flat-field imaging.

5.1 Lens Selection Rationale

Parameter	Value / Justification
L1 focal length (f_1)	75 mm — Thorlabs AC254-075-B (25.4 mm dia, NIR B-coat 650–1050 nm)
L2 focal length (f_2)	100 mm — Thorlabs AC254-100-B (25.4 mm dia, NIR B-coat 650–1050 nm)
4F magnification	$M_{4F} = f_2/f_1 = 100/75 = 1.333\times$
Total system magnification	$M_{\text{total}} = 0.476 \times 1.333 = 0.635\times$
Lens type	Achromatic doublets — eliminate chromatic aberration critical for NIR imaging

5.2 Image Size on Sensor (Verification)

Region	Final image diameter on sensor
Macula (5.5 mm retinal dia)	$5.5 \times 0.635 = 3.49 \text{ mm} \rightarrow 96\% \text{ of sensor height (3.63 mm)} \checkmark$
Full retina (32 mm)	$32 \times 0.635 = 20.3 \text{ mm} \text{ — larger than sensor (macular imaging mode)}$
Pixel resolution on retina	$1.4 \text{ } \mu\text{m} / 0.635 = 2.2 \text{ } \mu\text{m} \text{ per pixel on retina — resolves major retinal features}$

5.3 Distances Within the 4F Relay

Spacing	Distance
Aerial image plane $\rightarrow$ L1	$f_1 = 75 \text{ mm}$
L1 $\rightarrow$ L2 separation	$f_1 + f_2 = 75 + 100 = 175 \text{ mm}$
L2 $\rightarrow$ Sensor	$f_2 = 100 \text{ mm}$
Total 4F relay length	$2f_1 + 2f_2 = 150 + 200 = 350 \text{ mm}$
Full system length (cornea to sensor)	$10 + 35.71 + 75 + 175 + 100 = 395.7 \text{ mm} \approx 40 \text{ cm}$

6. LED Illumination Ring Design

6.1 Principle: Annular Illumination

The central challenge in retinal imaging is separating the illumination path from the imaging path to eliminate the bright corneal reflex. This is achieved by making the illumination annular: the LED ring projects light through the outer annulus of the pupil, while the camera collects light through the central pupil zone. The LED ring is co-planar with the condensing lens ($z = 10 \text{ mm}$ from cornea).

6.2 Ring Dimensions (Calculated)

Parameter	Value / Derivation
Camera collection beam diameter at condensing lens plane	$2 \times (\text{NA}_{\text{eye}} \times f_{\text{cond}}) = 2 \times (4/24) \times 35.71 = 11.9 \text{ mm}$
Ring inner diameter (clear aperture)	14.3 mm (1.2 $\times$ camera beam dia — prevents vignetting of imaging path)
Ring inner radius	7.1 mm
Ring outer diameter	20.3 mm (ring width = 3 mm)
Ring outer radius	10.1 mm
Angular range at cornea (from axis)	Inner: 35.5° Outer: 45.4° — both within dilated pupil acceptance ($\leq 53^\circ$) $\checkmark$
Vignetting condition	Ring inner clear aperture (14.3 mm) > camera beam (11.9 mm) — no vignetting $\checkmark$

Critical note from whiteboard: the LED ring hole (inner diameter) must be slightly smaller than the image size — this refers to ensuring the ring structure does not clip the imaging beam. Our 14.3 mm inner clear aperture vs. 11.9 mm imaging beam satisfies this with 20% margin.

6.3 LED Wavelength & Driver

- Wavelength: 850 nm NIR — minimal pupil constriction response, good IMX708 sensitivity, sufficient choroidal/retinal contrast
- Alternative: 780 nm for slightly better sensitivity on IMX708
- LED Driver: Use an off-the-shelf constant-current IC (e.g., AMC7135 at ~\$1 each, or Thorlabs LEDD1B for bench use). Buying a driver is strongly recommended over designing one — it saves time and ensures stable, flicker-free current regulation critical for consistent exposure.
- Exposure time: 1–10 ms recommended to freeze eye motion and retinal pulsation

7. Camera Focus Distance

The RPi Camera Module 3 has Phase Detection Autofocus. For this system, the 4F relay forms the final image exactly at the sensor plane, so the camera should be treated as receiving a beam from a virtual source at the back focal length of $L_2 = 100$ mm.

- Set AfMode to manual via libcamera: `--autofocus-mode manual`
- Set LensPosition to 0 (infinity focus) or tune via: `rpicam-still --autofocus-mode manual --lens-position 0`
- The reference paper's 8 cm focus distance can be matched by substituting $f_2 = 80$ mm ($L_2 = \text{AC254-080-B}$), adjusting total system length to 375.7 mm and reducing 4F magnification to $1.067\times$ (macula fills 77% of sensor height).
- Recommended: use $f_2 = 100$ mm for the 96% sensor fill factor unless the reference paper's exact configuration is required.

8. Wavelength Considerations & Lens Materials

8.1 Does the +28D Ophthalmoscopy Lens Work for NIR?

Standard clinical condensing lenses (Volk, Ocular, etc.) are made of optical glass (N-BK7 or equivalent) with broadband AR coatings. N-BK7 transmits well from 350 nm to beyond 2000 nm, so NIR transmission is not a concern. The key considerations are:

- Chromatic aberration: NIR light focuses slightly differently than visible. For a singlet condensing lens, this introduces a small focal shift — negligible at $f=35.7$ mm for the ~ 100 nm bandwidth of an 850 nm LED.
- AR coating: Verify the condensing lens AR coating covers 850 nm. Volk clinical lenses are optimized for visible; consider a broadband or NIR-specific coating if reflections are an issue.
- 4F achromats: Thorlabs AC254-xxx-B lenses have broadband AR coating optimized for 650–1050 nm — ideal for 850 nm imaging.

8.2 Liquid Lens Option (for myopia/hyperopia compensation)

As discussed: liquid lenses (e.g., Varioptic/Corning Arctic series) electrically tune focal length with high precision and fine step resolution. Placing one between the condensing lens and L1 allows real-time compensation for different refractive errors without mechanical movement. This is highly recommended

for clinical use but adds cost and control complexity. For the capstone prototype, a fixed condensing lens with manual WD adjustment is acceptable.

9. Recommended Component List

#	Component	Specification	Notes
1	Camera	RPi Camera Module 3 NoIR	NoIR variant REQUIRED for NIR imaging
2	Condensing lens	Volk 28D aspheric condensing lens	$f = 35.71$ mm, WD = 10 mm from cornea
3	4F Lens L1	Thorlabs AC254-075-B	$f = 75$ mm, 25.4 mm dia, 650–1050 nm coat
4	4F Lens L2	Thorlabs AC254-100-B	$f = 100$ mm, 25.4 mm dia, 650–1050 nm coat
5	NIR LED ring	850 nm, inner dia 14.3 mm, outer dia 20.3 mm	Custom PCB ring or Thorlabs ring illuminator
6	LED driver	AMC7135 constant current IC (~\$1) or Thorlabs LEDD1B	Buy, do not design — stable current essential
7	Eye phantom	Navarro model eye or equivalent	Fixed pupil ~6 mm; retina with test pattern

10. Open Questions for Supervisor

- Condensing lens AR coating: Should we verify 850 nm AR performance, or is a broadband clinical lens acceptable?
- Eye phantom pupil size: Fixed at 6 mm or should we test multiple sizes to simulate clinical variation?
- 4F lens aperture adequacy: 25.4 mm diameter lenses — confirm these are not undersized given 15.2 mm aerial image of full retina (may cause marginal vignetting at field edges).
- Reference paper 8 cm focus: Confirm whether $f_2 = 80$ mm or $f_2 = 100$ mm better matches your target specification.
- Liquid lens: Out of scope for v1 prototype, or worth including now for refractive error compensation?
- LED exposure sync: Should LED strobing be synchronized to camera frame timing, or continuous illumination is acceptable for v1?

Summary: Key Numbers

Parameter	Value
Total system length (cornea → sensor)	395.7 mm $\approx$ 40 cm

Total system magnification	0.635× (eye+cond: 0.476×, 4F: 1.333×)
Macula image on sensor	3.49 mm = 96% of sensor height
Retinal resolution (per pixel)	2.2 μm on retina per sensor pixel
LED ring inner diameter	14.3 mm (at condensing lens plane)
LED ring outer diameter	20.3 mm (at condensing lens plane)
NIR wavelength	850 nm
4F lens pair	$f_1 = 75$ mm (L1), $f_2 = 100$ mm (L2)
L1–L2 separation	175 mm
Camera variant	RPi Camera Module 3 NoIR (ESSENTIAL)